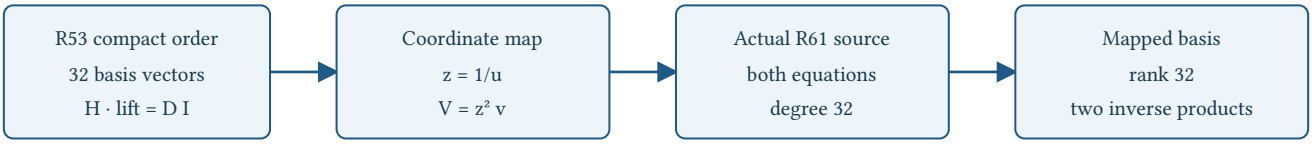
R64: compact-order bridge to the actual field

The retained rank-32 compact PID order is now bound to the actual R61 degree-32 source under $z = \frac{1}{u}$, $V = z^2 v = \frac{v}{u^2}$, and old $\lambda = l$. Both old source equations reduce to zero in the actual quotient. The source matrix identity $HM = DI$, where M is the saved source lift, holds before and after coefficient inversion. All 32 compact basis functions map to nonzero actual functions; their complete coefficient matrix has rank 32 and a checked two-sided inverse. Altered quadratic and matrix entries fail.



Full attempt record

Attempt	Status	Wall seconds	Error lines
bridge64-01	rejected	2	23
bridge64-02	rejected	1	22
bridge64-03	rejected	1	20
bridge64-04	accepted	26	0

The first attempt cast a source module after switching rings. The second mapped the module before casting, creating a zero fourth image. The third tested the original identity on that changed representation. All three are rejected despite late markers. Attempt 04 uses the original R53 matrix construction order and passes the strict gate. Wall intervals are receipts, not controlled performance comparisons.

Exact boundary and next step

The accepted run serializes both source equations, all 32 actual basis functions, the 32 by 32 image matrix and its inverse. Four frozen raw attempts and 24 historical byte pins are in results.json. The bridge identifies the generic field represented by the compact order. It does not certify that the order is integrally closed.

The next exact calculation starts with its trace dual. The retained Gram discriminant predicts a finite quotient of F2 dimension 160. At each descending Frobenius step, construct the complete binary square map on that quotient, solve its full kernel and reconstruct the PID lattice. Equality certifies affine normalization under the checked source hypotheses. Then calculate branch valuations on both proper charts for the level-77 map and subgroup action.

Primary normalization source: Singh–Swanson, Theorem 1.1.